Chapter 1

Compound Interest

When you loan money to someone, you typically charge them some sort of interest. The most common loan of this sort is what the bank calls a “savings account”. Any money you put in the account is loaned to the bank. The bank then lends it to someone else, who pays interest to the bank. The bank gives some of that interest to you. However, what if you leave the interest in your account? And you start making *interest on the interest*? This is known as *compound interest*.

1.1 An example with annual interest payments

Let’s say that you put $1000 in a savings account that pays 6% interest every year. How much money would you have after 12 years? Let’s make a spreadsheet.

	A	B	C
1	**Interest Rate**	6.00%	
2			
3	**After year:**	**Interest**	**Balance**
4	0	$0.00	1000
5	1	$60.00	$1,060.00

Create a new spreadsheet and edit the cells to look like this. All the cells in rows 1 - 4 are just values: just type in what you see.

The fifth row is all formulas:

After year	Interest	Balance
= A4 + 1	= B$1 * C4	= C4 + B5

The interest rate field should be formatted as a percentage. One thing to know when dealing with percentages in the spreadsheet: If the field says “600%”, its value is 6.

The cells in the Interest and Balance column should be formatted as currency.

You are about to make several copies of the cells in the fifth row, so make sure they look right.

Click on A5 and shift-click on C5 to select all three cells. Drag the lower-right corner down to fill the rows 6 - 15.

A5:C16 fx = A4 + 1

	A	B	C
1	**Interest Rate**	6.00%	
2			
3	**After year**	**Interest**	**Balance**
4	0	$0.00	1000
5	1	$60.00	$1,060.00
6	2	$63.60	$1,123.60
7	3	$67.42	$1,191.02
8	4	$71.46	$1,262.48
9	5	$75.75	$1,338.23
10	6	$80.29	$1,418.52
11	7	$85.11	$1,503.63
12	8	$90.22	$1,593.85
13	9	$95.63	$1,689.48
14	10	$101.37	$1,790.85
15	11	$107.45	$1,898.30
16	12	$113.90	$2,012.20
17			

Look at the numbers. The first interest payment is $60, but the last is $113.90. Your balance has more than doubled!

1.2 Exponential Growth

We figured this out numerically by repeatedly multiplying the balance by the interest rate. What if you wanted to know what the balance would be n years after investing P_0 dollars with an annual interest rate of r? (Note that r in our example would be 0.06, not 6.0.)

Each year, the balance is multiplied by $1+r$, so after one year, P_0 would become $P_0 \times (1+r)$. The next year you would multiply this number by $(1+r)$ again: $P_0 \times (1+r) \times (1+r)$. The next year? $P_0 \times (1+r) \times (1+r) \times (1+r)$ See the pattern? P_n is this balance after n years, then

$$P_n = P_0(1+r)^n \tag{1.1}$$

1.1 is referred to as the standard *compound interest formula*. Because n is an exponent, we call this *exponential growth*. There are few things as terrifying to a scientist as the phrase "The population is undergoing exponential growth".

If the principal is not compounded annually, we can modify the equation to account for that by changing n.

$$n = mt$$

Where t is the frequency of the compounding, called the period: (1: annually, 12: monthly, 52: weekly, 365: daily), and m is the number of periods. This equation becomes

$$P_n = P_0(1 + \frac{r}{n})^{nt}$$

We will see these two terms pop up in other areas of the curriculum like circular motion and waves.

1.3 Sensitivity to interest rate

For most people, the first surprising thing about compound interest is how quickly your money grows after a few years. The second thing that is surprising is how much difference a small change in the percentage rate makes.

Let's add another set of columns that shows what happens to your money if you convince the bank to pay you 8% instead of 6%.

Copy everything from columns B and C:

	A	B	C	D	E
1	**Interest Rate**	6.00%		6.00%	
2					
3	**After year**	**Interest**	**Balance**	**Interest**	**Balance**
4	0	$0.00	1000	$0.00	1000
5	1	$60.00	$1,060.00	$60.00	$1,060.00
6	2	$63.60	$1,123.60	$63.60	$1,123.60
7	3	$67.42	$1,191.02	$67.42	$1,191.02
8	4	$71.46	$1,262.48	$71.46	$1,262.48
9	5	$75.75	$1,338.23	$75.75	$1,338.23
10	6	$80.29	$1,418.52	$80.29	$1,418.52
11	7	$85.11	$1,503.63	$85.11	$1,503.63
12	8	$90.22	$1,593.85	$90.22	$1,593.85
13	9	$95.63	$1,689.48	$95.63	$1,689.48
14	10	$101.37	$1,790.85	$101.37	$1,790.85
15	11	$107.45	$1,898.30	$107.45	$1,898.30
16	12	$113.90	$2,012.20	$113.90	$2,012.20
17					

Now edit the second interest rate to be 8%:

	A	B	C	D	E
1	**Interest Rate**	6.00%		8.00%	
2					
3	**After year**	**Interest**	**Balance**	**Interest**	**Balance**
4	0	$0.00	1000	$0.00	1000
5	1	$60.00	$1,060.00	$80.00	$1,080.00
6	2	$63.60	$1,123.60	$86.40	$1,166.40
7	3	$67.42	$1,191.02	$93.31	$1,259.71
8	4	$71.46	$1,262.48	$100.78	$1,360.49
9	5	$75.75	$1,338.23	$108.84	$1,469.33
10	6	$80.29	$1,418.52	$117.55	$1,586.87
11	7	$85.11	$1,503.63	$126.95	$1,713.82
12	8	$90.22	$1,593.85	$137.11	$1,850.93
13	9	$95.63	$1,689.48	$148.07	$1,999.00
14	10	$101.37	$1,790.85	$159.92	$2,158.92
15	11	$107.45	$1,898.30	$172.71	$2,331.64
16	12	$113.90	$2,012.20	$186.53	$2,518.17
17					

1.4 Continuous Compound Interest

Let's say, hypothetically, that we compound interest on a given balance for an *indefinite* amount of time. Then, we take the formula

$$P_n = P_0(1 + \frac{r}{n})^{nt}$$

What if we divide our time period into infinite chunks? This would mean splitting our n into much smaller pieces. We can use calculus notation to find the limit of this:

$$\lim_{n\to\infty} P_0(1 + \tfrac{r}{n})^{nt}$$

we can create a variable $x = \frac{n}{r}$. We do this so that we eliminate some other dummy variables. Now we have

$$\lim_{n\to\infty} P_0(1 + \tfrac{1}{x})^{xrt}$$

as we can eliminate n with $n = xr$.

Now we can isolate parts of our compound interest equation into the limit, and extract variables out.

$$P_0 \left(\lim_{n\to\infty} (1 + \tfrac{1}{x})^x \right)^{rt}$$

Fun math fact: $\lim_{n\to\infty}(1 + \frac{1}{x})^x$ is equal to the mathematical constant e. We are going to dive into this in the *logs* chapter! For now, just understand that it is a constant in mathematics that describes many growth or decay values. With that, we can simplify the equation as the continuous compound interest equation (1.2)

$$P_0 e^{rt} \tag{1.2}$$

1.5 Practice Exercises

Exercise 1

You deposit \$1500 into a savings account that pays an annual interest rate of 5%, compounded annually. What is the balance after 8 years?

Note: you can also use excel or google sheets built in function `=FV()`.

Working Space

Answer on Page 7

Exercise 2

A bank offers two savings accounts, both with a stated annual interest rate of 6%. Account A compounds monthly ($t = 12$) and Account B compounds daily ($t = 365$). If you deposit \$3000 into each account, which account has the higher balance after 5 years, and by how much?

Create two columns in Excel/Sheets showing the balance after each year for each account, using \$3000 as the principal. Start with year 0 for the initial deposit, then fill in years 1 through 5 using the compound interest formula for monthly and daily compounding.

Working Space

Answer on Page 7

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 5)

Using 1.1 with $P_0 = 1500$, $r = 0.05$, and $n = 8$:

$$P_8 = 1500(1 + 0.05)^8 = 1500(1.05)^8 \approx \$2216.18$$

Alternatively, in Sheets/Excel, we can use `=FV(0.05, 8, 0, -1500)` to say that we want the *future value* of 1500 deposited (hence, negative), after 8 years compounded once per year.

Answer to Exercise 2 (on page 5)

Using $P_n = P_0(1 + \frac{r}{n})^{nt}$ with $P_0 = 3000$, $r = 0.06$, and $n = 5$ years:

Account A (monthly, $t = 12$):

$$P = 3000\left(1 + \frac{0.06}{12}\right)^{12\times5} \approx \$4046.55$$

Account B (daily, $t = 365$):

$$P = 3000\left(1 + \frac{0.06}{365}\right)^{365\times5} \approx \$4049.48$$

	A	B	C	D
1	**Principal**	$ 3,000.00		
2	**Annual Rate**	6%		
3				
4	**Year**	**Account A**	**Account B**	
5	0 - initial	$3,000	$3,000	
6	1	$3,185.03	$ 3,185.49	
7	2	$3,381.48	$ 3,382.46	
8	3	$3,590.04	$ 3,591.60	
9	4	$3,811.47	$ 3,813.67	
10	5	$4,046.55	$ 4,049.48	

Account B has the higher balance, by about \$2.93. This makes sense: the more frequently interest is compounded, the closer the balance gets to the continuous compounding limit, P_0e^{rt}.

Index